

Shashwat Singh

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RESEARCH INTERESTS

My research focuses on bio-inspired robots, mechanical intelligence, and locomotion in unstructured environments. I am particularly interested in how physical embodiment, morphological adaptation, and robot environment interaction can simplify sensing and control, enabling robust locomotion across terrestrial, aquatic, and aerial domains.

EDUCATION

Carnegie Mellon University, Pittsburgh, PA, USA Aug. 2023 – Present
Ph.D. in Robotics
Advisor: Prof. Zeynep Temel

University at Buffalo, Buffalo, NY, USA Aug. 2021 – Jul. 2023
M.S. in Robotics
Advisor: Prof. Ryan St. Pierre
Thesis: Buffalo Byte: A highly mobile and autonomous millirobot platform

University of Allahabad, Allahabad, UP, India Aug. 2017 – Jun. 2021
B.Tech. in Electronics and Communication

RESEARCH EXPERIENCE

Research Assistant, Carnegie Mellon University, Pittsburgh, PA, USA Aug. 2023 – Present
Zoom Lab, advised by Prof. Zeynep Temel

- Designed, fabricated, and controlled transformable robots that reconfigure their bodies to switch locomotion modes, embedding terrain adaptation in the mechanism rather than in control.
- Designed coupling interfaces pairing a soft vine robot with centimeter-scale rollers, combining their complementary strengths for search and rescue in confined spaces.
- Built onboard perception and control stacks fusing vision, proprioception, and contact force to sense terrain and trigger morphology changes in real time.

Research Assistant, University at Buffalo, Buffalo, NY, USA Aug. 2021 – Jul. 2023
Robot Form and Function Lab, advised by Prof. Ryan St. Pierre

- Designed and fabricated sub-cubic-inch autonomous millirobots with multi-modal locomotion.
- Developed embedded control software under strict size, weight, and power (SWaP) constraints, including real-time sensing, actuation, and communication on custom PCBs.
- Performed dynamic modeling, trajectory optimization, controller design, and experimental validation on hardware, bridging algorithm analysis with physical implementation.

Research Assistant, University of Allahabad, Allahabad, UP, India Jul. 2019 – Jun. 2021
Machine Vision and Intelligence Lab

- Built a handheld device enabling visually impaired students to perform science experiments; filed as patent.
- Partnered with NCERT (India's national body for school curricula) to bring the device across nation.

AWARDS

Best Poster Award – ICRA 2025 Amphibious Robot Workshop, “PuffyBot: An untethered shape morphing robot for multimodal locomotion.”

Best Poster Award – ICRA 2024 Unconventional Robot Workshop, “Multi-modal jumping and crawling in an autonomous, springtail-inspired microrobot.”

Dean’s Graduate Achievement Award – For excellence in research and technical publications.

PUBLICATIONS

Bold indicates lead/first authorship. Full list on [Google Scholar](#).

1. **S. Singh**, S. Zhang, S. Matonis, Z. Temel. “[Terraskipper: A centimeter-scale robot for multi-terrain skipping and crawling.](#)” *IEEE Intl. Conf. on Robotics and Automation (ICRA)*, 2026.
2. **S. Singh**, J. Ferlazzo, S. Wang, S. N. Boateng, M. Kramer, M. O. Martinez, F. Z. Temel, L. H. Blumenschein. “[Physical coupling for collaboration in heterogeneous robot teams.](#)” *IEEE Robotics and Automation Letters (RA-L)*, 2026.
3. T. Han, **S. Singh**, S. Patil, Z. Temel. “[From fold to function: Dynamic modeling and simulation-driven design of origami mechanisms.](#)” *IEEE RoboSoft*, 2026.
4. **S. Singh**, Z. Si, Z. Temel. “[PuffyBot: An untethered shape-morphing robot for multi-environment locomotion.](#)” *arXiv:2511.09885*, 2025.
5. S. Yi, **S. Singh**, A. Seo, R. St. Pierre, K. Sycara, Z. Temel. “[Reconfigurable robot swarms for terrain traversal with passive coupling mechanisms.](#)” *Autonomous Robots*, 2025.
6. **S. Singh**, R. Truax, R. St. Pierre. “[Buffalo Byte: A highly mobile and autonomous millirobot platform.](#)” *IEEE Robotics and Automation Letters (RA-L)*, 2024.
7. **S. Singh**, Z. Temel, R. St. Pierre. “[Multi-modal jumping and crawling in an autonomous, springtail-inspired microrobot.](#)” *IEEE ICRA*, 2024.
8. S. N. Boateng, **S. Singh**, et al. “[Heterogeneous collaboration: A new approach for search and rescue operations.](#)” *IEEE Intl. Symp. on Safety, Security, and Rescue Robotics (SSRR)*, 2024.
9. N. Skowronski, M. M. Pour, **S. Singh**, S. J. Longo, R. St. Pierre. “[Design and control of jumping microrobots with torque reversal latches.](#)” *Bioinspiration & Biomimetics*, 2024.

TEACHING EXPERIENCE

Teaching Assistant, Carnegie Mellon University

2025

- 15-482 Autonomous Agents (Fall 2025): led labs and mentored projects in perception, planning, and learning for autonomous robots.
- 16-235 Fantastic Robots and How to Fold Them (Spring 2025): mentored teams in mechanism design, fabrication, and control of origami-inspired robots.

LEADERSHIP

- **Graduate Student Delegate**, Robotics Department, CMU: Represent the largest graduate student group in departmental governance.
- **Member**, Robotics Institute Student Organization (RISO), CMU: Organize social and professional events for 400+ graduate students.

MENTORSHIP

Master's Student Mentoring:

- [Kiran Marques](#) – Carnegie Mellon University, M.S. in Mechanical Engineering, 2026 (*first author, submitted to SSRR 2026*)
- [Sidney Nimako-Boateng](#) – Carnegie Mellon University, M.S. in Robotics, 2025 (*first author, SSRR 2024*)

Undergraduate/Visiting Student Mentoring:

- [Tianhui \(Tiffany\) Han](#) – Carnegie Mellon University (*first author, RoboSoft 2026*)
- [Jasper Chapman](#) – Carnegie Mellon University, [SURF](#), Summer 2026
- [Jayden Chen](#) – Carnegie Mellon University, Research Assistant Spring 2026 & [SURF](#) Summer 2026
- [David Alvarez Gomez](#) – Carnegie Mellon University, [SURA](#), 2026
- [Jumpei Saito](#) – Keio University, [RISS](#), Summer 2026
- [Sheri Zhang](#) – Carnegie Mellon University, [SURA](#), Summer 2025 (*second author, ICRA 2026*)
- [Kwesi Nkromo](#) – Tuskegee University, [RISS](#), Summer 2025
- [Anson Li](#) – Carnegie Mellon University, [SURF](#), Summer 2025
- [Derek Peng](#) – Carnegie Mellon University, Research Assistant, Summer 2025

ACADEMIC SERVICES

Workshop Organizer:

- [3rd Workshop on Unconventional Robots](#), ICRA 2026, Vienna, Austria
- [2nd Workshop on Unconventional Robots](#), ICRA 2025, Atlanta, USA

Journal/Conference Reviewer: IEEE Robotics and Automation Letters (RA-L), IROS, RSS, RoboSoft

INVITED TALKS

- [3rd Workshop on Unconventional Robots](#), ICRA 2026, Vienna, Austria
- [Workshop on Multi-Stable and Origami-based Soft Robots](#), RoboSoft 2026, Kanazawa, Japan

TECHNICAL SKILLS

Machine Learning & AI: Reinforcement Learning (PPO, SAC), Deep Learning, Machine Learning, Imitation Learning, Sim-to-Real Transfer, Optimal Control, PyTorch, TensorFlow

Robotics: Perception, Computer Vision, Multi-Modal Sensor Fusion, Controls, Dynamic Modeling

Programming & Simulation: Python, C++, ROS/ROS2, NVIDIA Isaac Lab/Isaac Sim, MuJoCo

Embedded & Hardware: NVIDIA Jetson Orin, Raspberry Pi, Arduino/microcontrollers, RS-485/Serial, CAN, sensor integration, PCB design (KiCAD)

Mechanical Design & Fabrication: SolidWorks, Fusion 360, OnShape, CNC machining, rapid prototyping